A Lightweight Multimodal Fusion Framework for Multi-Label Document Image Quality Assessment⋆

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ABSTRACT

The image quality of documents is critical in automated document processing systems, particularly those involving Optical Character Recognition (OCR) for text extraction. When documents are scanned or captured, they may suffer from defects such as blur, skew, or poor lighting, which degrade text extraction performance. This makes Document Image Quality Assessment (DIQA) an essential component of document processing pipelines. Traditional DIQA models rely on a single input modality and produce a single overall quality score, limiting their usefulness in real-world applications. To address this limitation, we propose a lightweight multimodal framework for multi-label document defect detection. The proposed approach combines visual features learned from images using a lightweight Convolutional Neural Network (CNN) with handcrafted numerical features that describe statistical properties of document images. Experimental results show that the proposed approach achieves a Macro F1-score of 0.9154, demonstrating strong performance in multilabel defect detection. In addition, the proposed model exhibits stable training behaviour and avoids overfitting compared to pre-trained models like MobileNet, EfficientNet, and ResNet-18. Furthermore, the framework is computationally efficient, with a model size of only 1.44 MB, high throughput, and low latency, making it suitable for real-time document processing in low-resource environments.

1. Introduction

The rapid growth of automated document processing systems relies heavily on OCR to extract information from document images in applications such as invoice processing, archiving, and identity verification (Know your customer - KYC) verification, where the accuracy of the extracted information heavily depends on image quality. Therefore, DIQA before OCR has become an essential step in the document processing pipeline because an efficient DIQA system can act like a gatekeeper who rejects poor-quality images at the earliest stage of the pipeline.

Despite improvements in camera technology, document images captured in mobile environments often suffer from defects such as motion blur, out-of-focus capture, document skew, glare, and other lighting issues. The existing literature on DIQA typically formulates this task as predicting a single quality score or a binary classification. However, a single score or a binary classification indicates overall quality but fails to reveal the nature of the defect. This limitation prevents downstream processing like taking specific actions to correct the defect. It also prevents providing feedback to the user in an interactive system such as a KYC system (e.g., asking the user to hold the camera steadier or improve lighting). To provide such feedback in real-time, it is necessary to design a system that can identify multiple types of defects within a document image.

To develop such quality assessments, previous approaches in DIQA have generally followed different routes with their own limitations. For example, the earlier methods focused on using handcrafted numerical features to estimate a quality score such as the character gradient method proposed by Li, H., et al., (2018) (2) and the foreground contrast proposed by Alaei, Alireza, et al., (2015) (3). The disadvantage of these techniques is that, even though they are computationally efficient, they do not actually understand the context of the image. To address this issue deep learning models such as CNNs were introduced to capture and understand the visual context. However, deep learning approaches alone still exhibit limitations. These approaches have different objectives. For example, Shen et al. (2020) (15) and Wenniger et al., (2023) (16) incorporated visual and textual data to assess the semantic quality of the documents and not visual quality. Whereas IQA-GL (Wang et al., 2022) (14) or DocIQ (Ma, Zhichao, et al., 2025) (13) use complex visual mixing to predict visual quality. However, they only provide a simple score. On the other hand, frameworks such as DeQA-Doc (Gao et al., 2025) (9) or Ocean-OCR (Chen et al., 2025) (10) use Large Language Models (LLMs) achieving high accuracy, but at significant computational cost. This highlights the need for a model that clearly identifies multiple specific defects in a document in a computationally efficient manner.

To address this gap, we propose a lightweight multimodal framework which detects multi-label document defects. The proposed framework combines visual features learned by a CNN with numerical features which describe the statistical aspects of the defects. Rather than using computationally expensive fusion techniques, we use a simple projection layer to combine what the CNN "sees" with the mathematical features. This combination will address the

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limitations of using handcrafted features and deep learning models alone and enables the system to achieve improved accuracy in classifying multiple defects. Evaluation of this framework on real-world SmartDoc-QA dataset demonstrates competitive performance when compared to existing, and achieves strong performance without the use of any attention-based architectures. Also, when compared to popular lightweight pre-trained models such as MobileNet, EfficientNet, and ResNet-18 our model performs more consistently on new data, while the pre-trained models suffered severe overfitting. **The main contributions of this work are:**

- Frames DIQA as a multi-label defect classification problem, enabling the model to detect multiple defects simultaneously which helps in providing targeted feedback for improving OCR performance.
- Proposes a lightweight hybrid multimodal framework that combines visual features from images with statistical features, using a balanced fusion approach so both the modalities contribute effectively.
- Implementation of a computationally efficient fusion framework with low computational requirement, making it suitable for fast and efficient use in low-resource environments.
- Demonstrating the framework's ability to achieve competitive performance compared to existing models and standard attention mechanisms, and overcome severe overfitting prevalent in pre-trained architectures.

2. Related Work

This section reviews existing DIQA methods, from the early handcrafted numerical models to visually aware deep learning models to the use of Large Language model and multimodal techniques.

2.1. Handcrafted Numerical Methods

The early DIQA methods used direct mathematical formulas to assess the quality of the document images. These approaches were based on the assumption that the quality of the text in the images can be measured using mathematical numerical features such as sharp edge gradients. A popular example is the Character Gradient-DIQA (CG-DIQA) system proposed by Li, H., et al., (2018) (2) using the Maximally Stable Extremal Region (MSER) algorithm. Then it measures sharpness of the text using Sobel filters and determines the probability of a successful OCR. Similarly, Alaei, Alireza, et al., (2015) (3) proposed a model based on Gradient Magnitude Similarty Deviation (MGMSD). This model separates the foreground text and background and allows the model to measure the severity of the defects while ignoring background noise.

Even though these mathematical techniques are extremely fast, a study by Kopytek et al., (2024) (4) showed that mathematical techniques such as Peak Signal-to-Noise Ratio (PSNR) (5) and Pseudo-F-Measure (6) achieved high accuracy on lab-generated data, but the accuracy dropped to 0.25 and 0.57 when tested on real world images.This drop in performance occurs because mathematical formulas lack visual understanding, and cannot differentiate between a blurry background and a blurry word. However, due to their efficiency in calculating the severity of defects these formulas were not abandoned. They were used along with visual cues using deep learning models.

2.2. Deep Learning-Based Approaches

To overcome the limitations of mathematical formulas the field moved to deep neural network based models. One of the early examples is DeepBIQ model proposed by Li, Jie, et al., (2016) (7). This model uses a CNN to extract visual features and Support Vector Regression (SVR) to predict a single image quality score.

After this, various complex architectures were created to address document-related challenges. One such approach is the attention-based Recurrent Neural Network (RNN) framework proposed by Li, Pengchao, et al., (2017) (8). This model uses an attention mechanism to identify text regions and process them sequentially using CNN and Gated Recurrent Unit (GRU) which together create a summary of the document quality. This framework performed well on generated dataset but showed reduced accuracy when tested on a real-world dataset like SmartDoc-QA (Nayef et al., 2015) (1). This model was computationally expensive as each text region was processed sequentially.

The deep learning models significantly improved performance in DIQA when compared to handcrafted features. Even though some models like DeepBIQ perform well and can be used in low resource settings, many architectures such as the RNN framework proposed by Li, Pengchao, et al., (2017) (8) are computationally expensive and cannot be used in real world scenarios. However, these deep learning models usually predict only a single quality score and do not indicate the specific defects in the document image.

2.3. Large Language Model-Based Approaches

Recently, research in the field of DIQA has shifted towards exploring Large Language Models (LLMs) to achieve high accuracy. For example, the DeQA-Doc framework proposed by Gao et al., (2025) (9) utilises LLMs capable of processing high resolution images. It achieved a high performance on DIQA – 5000 dataset. Another example, is the Ocean-OCR framework proposed by Chen et al., (2025) (10) which used a large-scale model. This model avoids shrinking the images and hence captures minute details and outperforms professional text reading software.

However, this large-scale model approach achieves high accuracy but has a major drawback. These models are massive and are computationally very expensive and deployment of these models requires powerful servers. Moreover, despite the massive size of these models, they struggle with certain document problems. The OCRBench study by Liu et al., (2023) (11) demonstrated that even advanced models struggle when handling blurry text or random numbers. This is because they are designed for language understanding and do not effectively understanding visual features. Finally, the Q-Doc benchmarking by Huang et al., (2025) (12) revealed that to make these LLMs score the quality of an image, the models are required to “think out loud” step by step before it gives an answer. This extra thinking step could make the system even slower. This shows the need for a lightweight, time and memory efficient model that can be used in even low-resource environments.

2.4. Multimodal and Fusion Approaches

To overcome the limitations of single modality models, multimodal feature fusion techniques have been explored. Multimodal feature fusion in DIQA is performed by combining different modalities of information such as visual features and numerical features to help models learn complex defect patterns. The literature survey shows that this feature fusion is performed for two different reasons. These methods are used to assess the semantic quality or the visual quality of the document image.

In semantic quality assessment, the models combine textual information and image layouts to assess the quality of the content in the document. For example, Shen et al., (2020) (15) combines CNN and RNN streams to assess the semantic quality of academic papers while MultiSChuBERT (Wenniger et al., 2023) (16) combined text and layout information to predict document impact. On the other hand, there are models built for visual quality assessment that combine multiple visual features with mathematical features to improve defect detection. For example, the AHIQ framework (Lao et al., 2022) (17) combines global Vision Transformer (ViT) with a local CNN. Similarly, the DocIQ model (Ma, Zhichao, et al., 2025) (13) fuses the structural layout of the page with the features learned by a CNN. This allows the model to focus on text regions, including tables and figures for defects, rather than blank spaces. Other hybrid models (Alaei, Alireza, et al., 2024) (18) combine deep visual features with mathematical features.

The standard output of the vast majority of the visual quality assessment fusion models is a single quality score. Another technical challenge that exists when it comes to fusion of modalities is how we mix them together while making sure the features complement each other and not overpower one another. Usually, fusion is done by concatenating the features. However, Wang et al., (2022) (14) explained in IQA-GL model, concatenation leads to a problem known as “modality drowning”. This happens because the features learned by deep learning models are high-dimensional and complex and may drown out the small numerical features. To fix this IQA-GL used a technique called “Bilinear Pooling” which multiplies the features together in a complex mathematical combination. Even though this technique prevents feature drowning, it is extremely memory and computationally expensive.

In summary, past research shows significant progress in the field of DIQA, with each approach improving and contributing in different ways. While handcrafted numerical features are fast, they struggle when used alone on real- world images but remain efficient in measuring defect severity. Deep learning models introduced the visual understanding which the numerical models lacked and improved accuracy in identifying defects. Large Language Models improve accuracy further but they are computationally expensive to use with a risk of hallucination and unreliable predictions. Combining these approaches through multimodal fusion has shown that one modality can fix the shortcomings of the other. However, the vast majority of these models output a single quality score or a binary classification, which highlights the need for multi-label defect detection for downstream applications. Additionally, when building multimodal fusion models the review has clearly highlighted the need to choose fusion techniques that ensure balance of modalities in an efficient manner. This highlights the need for a framework that is fast, predicts multi-label outputs by combining visual and numerical features in a balanced manner which is computationally inexpensive and suitable for deployment in low-resource environments. The design and implementation of this proposed framework will be detailed in the following section.

3. Methodology

This section describes the proposed multimodal framework for document image quality assessment.

3.1. Dataset

In this work, we utilized the SmartDoc-QA dataset (Nayef et al., 2015) (1), a well-know public dataset specifically created for DIQA. This dataset contains exactly 4260 real-world images of documents that were captured using smartphones in uncontrolled, realistic environments. The dataset is built from 30 single-page documents that includes modern report, old letters and store receipts. Each of the 30 documents was captured with different defects such as motion blur, focus blur, uneven lighting, shadows, glare and document skew and combinations of these defects resulting in a total of 4260 images. Unlike lab-generated datasets, that have artificial synthetic defects applied this dataset provides real-world perspective which is important when we want our model to be used in real-world scenarios.

3.2. Preprocessing

A preprocessing pipeline was applied before training. The dataset was collected under a controlled capture setting, where groups of images were taken from fixed camera angles and distances against a red background. Cropping was necessary because the images contain a red background, as shown in Figure 1, which distracts from the document. Due to the consistent capture setup, cropping regions were manually determined for each image group and applied to the entire dataset. The goal of cropping was to ensure that the document occupies most of the image while preserving visual defects.After cropping, all images were resized to 224 × 224 pixels for the neural network. Instead of stretching the

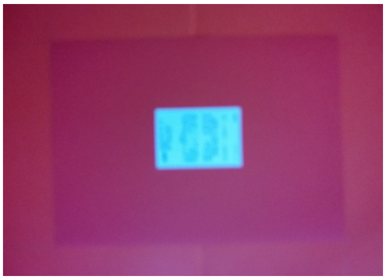


Figure 1: Sample original document image.

Figure 2: Sample original document image.

image, resizing was done by shrinking the image until its longest side was 224 pixels and padding was applied to fill the remaining empty space, producing a final square image of 224×224 pixels as seen in Figure 2

3.3. Defect Categorization

One of the key factors of the proposed framework is multi-label output, which addresses the limitations of traditional single score DIQA models. Images can contain either a single defect or multiple defects within the same document, and this is captured by the SmartDoc-QA dataset (Nayef et al., 2015) (1). The framework assesses images across three distinct categories:

1. **Blur Presence (blur_present):** This category covers images affected by blur, which causes the loss of sharpness. In this study, both motion blur and out-of-focus blur were combined into a single category to simplify the classification task and improve model reliability.
2. **Lighting (lighting):** This category covers images with glare issues, imbalance in lighting, or shadow obstructions.
3. **Document Skew (document_skew):** This category identifies the presence of perspective distortion or physical rotation of the document.

3.4. Multimodal Fusion Framework

We propose a lightweight, multimodal fusion framework that handles different types of image quality defects in Document images. Traditional models rely on either deep learning or handcrafted numerical features, which can lead to information blind spots. To address this, the proposed framework, as shown in Figure 3, combines both approaches in a dual-branch structure. One branch calculates the numerical features that measure image properties, while the other branch uses a neural network to learn visual patterns directly from the image. These two branches operate in parallel and their outputs are combined in a balanced projection layer to produce the final multi-label classification output.

3.4.1. Handcrafted Numerical Modality

As shown in the numerical modality branch of the architecture in Figure 3 the model first computes a set of numerical features that capture the measurable properties of the document image. These numerical features do the important job of quantifying severity or physical properties of image defects such as blur intensity or geometric distortions. Although CNNs are efficient in learning visual patterns, combining numerical features can complement the features learned by the CNN and enrich the model's knowledge of the defects.

Before feature extraction, each document image is converted into a grayscale image $I(x, y)$.

1. **Variance of Laplacian:** This features measures the sharpness of an image by calculating how quickly pixel intensity changes from one edge to the other. Sharp images will have higher variance as the text edges are sharp, while blurry images has smoother text edges resulting with lower variance.

$$F_{lap} = \mathrm{Var}(\nabla^2 I(x, y)) \tag{1}$$

2. **Tenegrad Measure:** This feature evaluates image sharpness by using the Sobel operator to compute image gradients along both the horizontal (G(x)) and vertical (G(y)) axes. High quality images with strong edges will produce larger gradient value, while blurry images with weak edges will have smaller gradient values.

$$F_{ten} = \frac{1}{N} \sum_x \sum_y \left(G_x^2 + G_y^2 \right) \tag{2}$$

3. **Brenner Gradient:** This feature measures intensity differences between one pixel and another pixel which is located two positions away from it in the horizontal direction. In sharp images, the pixel difference tends to be larger where low quality image produce smaller difference.

$$F_{bren} = \sum_x \sum_y (I(x+2, y) - I(x, y))^2 \tag{3}$$

4. **FFT High Frequency Ratio:** This feature uses Fast Fourier Transform to analyse the image in the frequency domain, where sharp images with sharp text edges have high frequency and blurry images that has smoother text edges have lower frequency.

$$F_{fft} = \frac{\sum_{(u,v)\notin R} |F(u, v)|}{\sum_{(u,v)} |F(u, v)| + \epsilon} \tag{4}$$

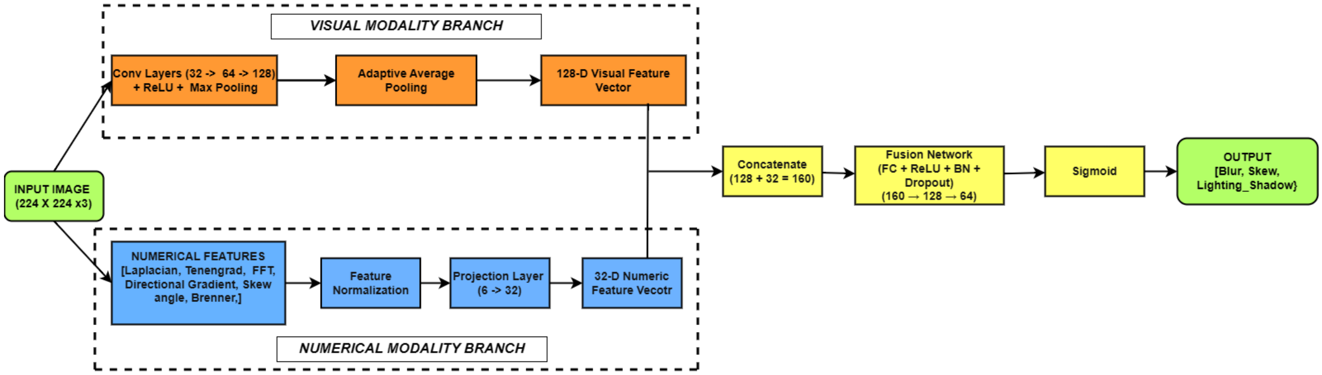


Figure 3: Proposed multimodal framework architecture.

5. **Directional Gradient Ratio:** This feature help to identify directional distortions and lighting variations by comparing the average strength of horizontal gradients with vertical gradients. Large differences between the two directions may indicate issues such as skew or uneven lighting.

$$F_{dir} = \frac{\mu(|G_x|)}{\mu(|G_y|) + \epsilon} \tag{5}$$

6. **Skew Angle via Hough Transform:** This feature estimates document rotation by detecting straight lines within a document image using Hough Transform.

$$F_{skew} = \frac{1}{K} \sum_{i=1}^{K} \left(\theta_i - \frac{\pi}{2} \right) \tag{6}$$

If we look at all the six handcrafted features, these features capture edge strength, sharpness, and pixel intensity, which are closely related to blur detection. However, they also help identify lighting-related issues like glare and shadow, which affect edge clarity in a similar manner. In order to include the effects of geometric distortion such as skew, Hough transform is used to measure the orientation of the prominent lines in the document. Once the six numerical features are computed, they are normalized using Z-score standardization to ensure consistent scaling. The feature vector is then passed through a fully connected layer followed by a ReLU activation function to learn a high-level representation of the numerical features. This transformed feature vector is then passed through the fusion stage where it is combined with the visual features learned from the CNN branch.

3.4.2. *Visual Feature Modality*

Handcrafted features provide numerical measures of defects but do not capture the visual structure or location of these defects that occur within the document image. To address this limitation, a CNN is used to learn the visual patterns directly from the image. To ensure the proposed model is suitable for deployment in low-resource environments, a lightweight CNN is used instead of using large, computationally expensive models.

After preprocessing of the raw images, the input image of size 224×224 is passed through three convolutional layers, where the feature maps increase from 32 to 64 and then to 128. Each layer uses a 3×3 filter and is followed by a ReLU activation function to learn complex visual patterns from the document image.

After the first two convolutional layers, a 2×2 max pooling layer is used to reduce spatial dimension and make the model focus only on important features. After the final convolutional layer, adaptive average pooling is applied to produce a compact feature vector of size 128, which represents the visual characteristics of the document image. Therefore, this visual feature vector captures important information about document structure, text layouts and defect patterns and is combined with the numerical feature vector in the fusion stage for defect prediction.

3.4.3. *Fusion Strategy*

An important step in a multimodal architecture is the fusion layer, where the goal is to ensure that different feature types are combined in such that they contribute efficiently. In our framework, the CNN branch produces a 128–dimensional feature vector, while the numerical feature branch produces only a 6-dimensional feature vector. If these feature vectors are combined directly, the model may tend to rely more on high-dimensional visual features leading to feature drowning, where the contribution of numerical feature is underrepresented.

To address this, the 6-dimensional numerical feature vector is passed through a learnable projection layer, where the dimension of the numerical feature is increased. It is first expanded to a 32-dimensional vector using a fully connected layer, followed by a ReLU activation function, batch normalization and dropout. This transformed vector is then passed through a second linear layer which further refines the representation. This ensures that the numerical feature vector has sufficient information and has enough representation to contribute.

Once both branches are processed, the 128-dimensional visual feature vector and 32-dimensional numerical feature vector are concatenated and are represented as:

$$x = [x_{\text{img}}, x_{\text{num}}] \tag{7}$$

A simple approach would be to pass the combination of the vectors through a single linear layer:

$$y = Wx + b \tag{8}$$

However, this captures only linear relationships and does not allow interaction between the modalities which helps capture complex patterns. To address this, a non-linear fusion block is used with a fully connected layer with ReLU activation, batch normalization and dropout. The fusion process is defined as:

$$y = W_3\, \sigma\left(W_2\, \sigma(W_1 x)\right) \tag{9}$$

where σ denotes the ReLU activation function.

This allows interaction between modalities and allows the model to learn complex patterns and relationships between the features. This concatenated 160-dimensional feature vector (128 visual + 32 numerical) is first projected into a 128–dimensional space, followed by ReLU activation, batch normalization and dropout for stable learning.

The representation is further reduced to a 64-dimensional feature vector using another fully connected layer to help the model focus on the most relevant information while reducing redundant information. Finally, the network outputs a 3-dimensional vector corresponding to the three defect categories: blur presence, document skew and lighting/shadow. This 3-dimensional vector is passed through a sigmoid function, converting each output into a probability score for each class, which is used to predict the presence or absence of each defect.

4. Experiments and Results

This section presents the evaluation of the proposed hybrid multimodal, multi-label classification framework on the SmartDoc-QA dataset (Nayef et al., 2015) (1).

4.1. Experimental Setup

The SmartDoc-QA dataset (Nayef et al., 2015) (1) contains document images with defects such as blur, skew and lighting issues, and was divided into training and validation sets, where 80% of the data was used for training and 20% for validation.

Preprocessing was performed where all images were resized to 224 × 224 and normalized before training. The model was trained using the *BCEWithLogitsLoss* function with the Adam optimizer (1×10^{-4}) and training was carried out for 30 epochs with a batch size of 16. F1-score is used as the primary metric of model performance for both class-wise and over all evalution due to its ability to balance precision and recall. Macro F1-score was used to measure performance across classes, while Micro F1-score evaluates performance across all sample document images. Accuracy was also calculated for each class to provide additional insight into the performance and behaviour of the model.

4.2. Baseline Models

Baseline models for individual modalities were evaluated to understand the effectiveness of the different features, a numerical-only neural network and an image-only model using CNN were built for this purpose. The numerical-only baseline uses handcrafted features to describe the statistical properties of an image, but it does not have the spatial understanding and cannot locate defects. In the image-only baseline, a CNN is used to learn visual pattern directly from the document images and is good at learning spatial and structural information but may not capture severity of the defects. The performance of the baseline models are summarised in Table 1.

Table 1
F1-scores of baseline models

Model	Blur	Skew	Lighting	Macro F1	Micro F1
Numerical-only baseline	0.8732	0.8133	0.4615	0.7160	0.7951
Image-only baseline	0.8634	0.8000	0.9188	0.8607	0.8607

As shown in Table 1, the image-only model performs better on lighting defects, while the numerical-only model performs better on blur and skew. This shows that the visual features are effective in capturing spatial information and numerical features are effective in measuring statistical properties. This observation highlights that both modalities have their own strengths and neither is sufficient at detecting document defect. This motivates the need for a hybrid approach that combines the spatial awareness of the visual features and statistical knowledge of the numerical features.

4.3. Hybrid Model

To overcome the limitations of single modality models, the hybrid approach was introduced by combining visual and numerical features, which allows the model to have both spatial understanding and statistical knowledge of the different defects. In this hybrid model, the 128-dimensional visual feature vector is concatenated with the 6-dimensional numerical feature vector and passed through a linear layer which captures only linear relationship between the features.

Table 2
Performance of hybrid model compared to baseline models

Model	Blur	Skew	Lighting	Macro F1	Micro F1
Numerical-only baseline	0.8732	0.8133	0.4615	0.7160	0.7951
Image-only baseline	0.8634	0.8000	0.9188	0.8607	0.8607
Basic Hybrid (Naïve Fusion)	**0.8978**	**0.8012**	**0.9518**	**0.8836**	**0.8729**

As shown in Table 2, the hybrid model outperforms both the baselines across all defect categories, achieving an overall improvement in Macro and Micro F1-scores. This shows that combining the visual and numerical features helped the model capture both spatial patterns and statistical properties of the document defects effectively. However, the performance of document skew slightly decreased when compared to the numerical-only model, which suggests that skew-related information captured by the numerical features are not fully utilized. This highlights the issue of feature drowning, where higher-dimensional feature dominates the lower-dimensional feature when they are directly concatenated. As a result, the model captures limited interaction between, motivating the need for an effective fusion strategy were both modalities can be integrated in a balanced manner.

4.4. Hybrid Model with Enhanced Fusion

To address the limitations observed in the hybrid model, a projection-based non-linear fusion strategy was introduced, as described in Section 3.4.3 where fusion is performed in a balanced and non-linear environment. This strategy ensures that the different modalities have equal representation and the model learns complex interactions between the visual and numerical features for defect detection. The performance of the fusion strategy is shown in Table 3

Table 3
Performance of enhanced fusion model compared to baseline and hybrid models

Model	Blur	Skew	Lighting	Macro F1	Micro F1
Numerical-only baseline	0.8732	0.8133	0.4615	0.7160	0.7951
Image-only baseline	0.8634	0.8000	0.9188	0.8607	0.8607
Basic Hybrid (Naïve Fusion)	0.8978	0.8012	0.9518	0.8836	0.8729
Fusion-Enhanced Hybrid Model	**0.9423**	**0.8228**	**0.9811**	**0.9154**	**0.8975**

The increases in F1-score across defect classes and document images, as shown in Table 3, indicates that the projection-based, non-linear fusion strategy helped the model to combine information from both modalities in an efficient manner. The improvement is particularly noticeable in blur and lighting detection, which suggests that the model is now able to efficiently integrate statistical and spatial information. In addition, document skew detection also improved, indicating that the earlier imbalance between visual and numerical features is effectively addressed. These results show that balanced interaction between modalities plays an important role in improving the performance of defect detection in document images.

4.4.1. Impact of Attention Mechanisms

To better understand and justify the design of our hybrid model, we conducted additional experiments by adding commonly used attention-based architectures to the visual branch. The goal is to check if explicitly guiding the models focus helped in increasing the models performance in identifying the defects when compared to our current approach. For this, we used the Squeeze-and-Excitation (SE) block and the Convolutional Block Attention Module (CBAM) as they help the model decide which features are important and where to focus in the image to detect them.

The SE block focuses on identifying which feature channels are more important by learning weights for each channel, which allows the model to focus on the useful. CBAM builds on this idea by not only deciding which features are important but also where in the image the model should focus for these features.

Table 4
Performance comparison of the proposed hybrid model with attention-based variants

Model	Blur	Skew	Lighting	Macro F1	Micro F1
Fusion-Enhanced Hybrid + SE block	0.9239	0.8117	0.9832	0.9063	0.8867
Fusion-Enhanced Hybrid + CBAM	0.9147	0.8184	0.9887	0.9073	0.8869
Fusion-Enhanced Hybrid Model	**0.9423**	**0.8228**	**0.9811**	**0.9154**	**0.8975**

Table 4 shows, that adding attention mechanisms did not lead to an improvement in the performance of the proposed model and that the proposed model without any attention blocks achieved better overall results across metrics. This indicates that the defects in document images are already effectively captured through the combination of the visual and statistical features, and explicitly guiding the models focus through channel or spatial weighting does not provide additional benefits. It also suggests that increasing model complexity with additional modules is not always beneficial and a simple, well-balanced design can perform better while remaining more efficient and give stable results in any environment.

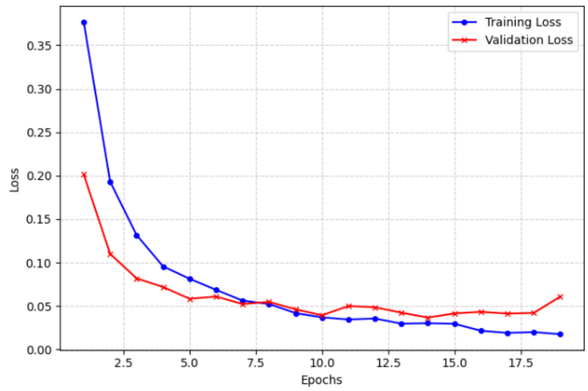
4.4.2. Comparison with Pre-Trained Deep Learning Models

Since pre-trained models are trained on large-scale datasets, a common approach in computer vision is to use these models and fine-tune them for a specific task. To evaluate the performance of our lightweight hybrid model, we tested by integrating well-known pre-trained models such as EfficientNet-B0, ResNet-18 and MobileNet into the visual branch of our framework.

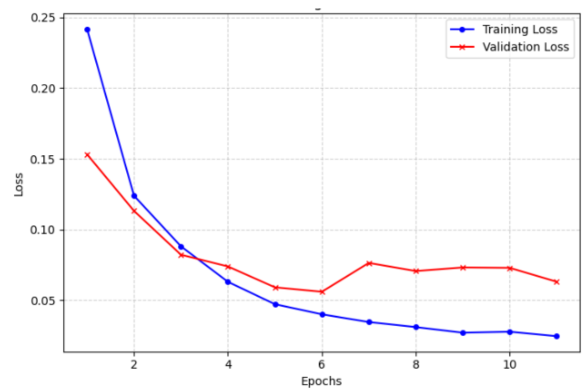
Initially, these models were trained for a 30 epochs , where overfitting was observed in the form of unstable validation loss curve. To ensure a fair comparison, early stopping was applied based on the validation performance, enabling evaluation under a more stable training setup.

Table 5 shows that these pre-trained models achieve very high performance across categories, with near-perfect scores, especially for lighting suggest possible overfitting. To better understand this, we examine how these models behave during training and validation.

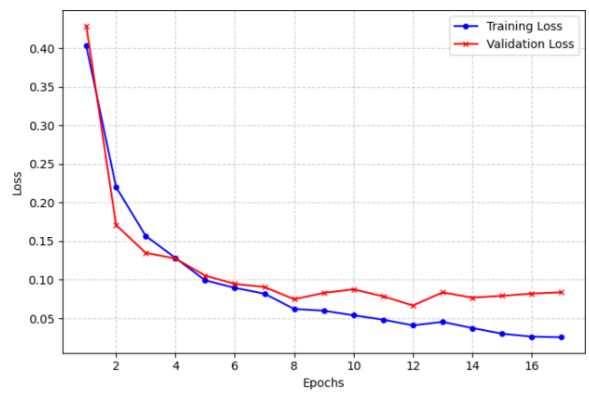
However, even after early stopping, all the pre-trained models show fluctuations in validation loss while training



(a) EfficientNet-B0



(b) ResNet-18



(c) MobileNet

Figure 4: Training and validation loss curves of pre-trained models: (a) EfficientNet-B0, (b) ResNet-18, and (c) MobileNet.

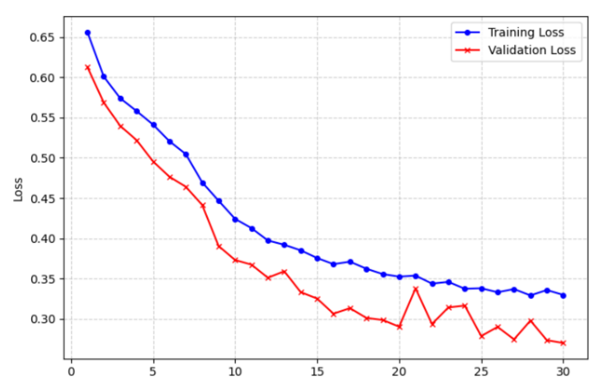


Figure 5: Training and validation loss curves of the proposed hybrid model.

Table 5
Performance of pre-trained models

Model	Blur	Skew	Lighting	Macro F1	Micro F1
EfficientNet-B0	0.9779	0.9903	1.0000	0.9894	0.9869
ResNet-18	0.9697	0.9843	1.0000	0.9847	0.9811
MobileNet	0.9755	0.9730	1.0000	0.9786	0.9743

loss is decreasing smoothly which indicates poor generalization. For EfficientNet-BO and ResNet-18, the validation loss begins to fluctuate after a few epochs and stops following the training trend, suggesting overfitting. This issue is most evident in Figure 4(c) where MobileNet displays the most unstable behaviour of all the three models, with noticeable fluctuations in validation. In contrast, Figure 5 shows that the proposed hybrid model shows smooth and stable training behaviour, where validation loss closely follows the training trend, indicating that the model is learning meaningful patterns and does not overfit.

Overall, while the pre-trained models appear strong based on their final scores, their training reveals that these

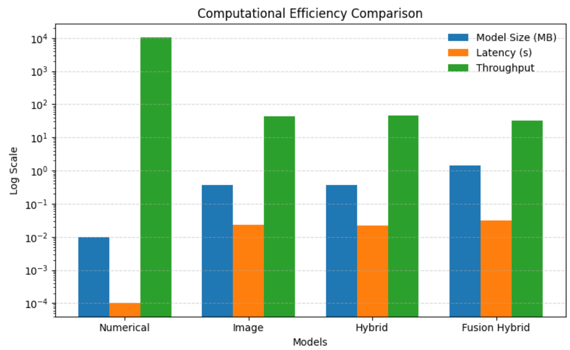


Figure 6: Comparison of computational efficiency metrics across models.

models trained on large-scale dataset easily memorize the specific defects in the training images instead of learning the underlying pattern. Meanwhile, the proposed lightweight hybrid framework avoids this trap and provides more stable and reliable performance by learning the underlying knowledge and properly applying it on unseen document images.

4.4.3. Computational Efficiency And Deployment Analysis

In addition to evaluating the performance of the proposed model, it is important to analyse the computational efficiency of the proposed framework to determine if it is suitable to deploy in real-world low-resource environments. Therefore, metrics such as model size, number of parameters were evaluated to understand the memory requirements of the model, while latency and throughput were measured to assess the time taken to process.

Table 6
Computational efficiency of the models

Model	Model Size (MB)	Latency (1 image)	Throughput (images/sec)
Numerical-only baseline	0.01	0.0001 s	**10603.73**
Image-only baseline	0.36	0.0229 s	43.60
Basic Hybrid model (Naïve Fusion)	0.36	0.0218 s	45.93
Fusion-Enhanced Hybrid Model	**1.44**	0.0317 s	31.57

From Figure 6, it can be observed that the numerical-only, image-only model and the standard hybrid model exhibit smaller model size, latency and throughput, but these models have their own weaknesses which makes them unsuitable as standalone solutions. Even though, the proposed hybrid model has an increased number of parameters and model size due to the additional layers for feature projection and fusion, with a slight increase in latency and reduction in throughput, it is still relatively small and suitable for low-resource environments. Overall, the results show that high detection performance can be achieved without relying on computationally expensive models, and that a carefully designed lightweight framework can provide both efficiency and reliability.

4.5. Comparison with Existing Work

To validate the effectiveness of the proposed framework, its performance was compared with the existing methodologies in the field of DIQA. By looking at the broad range of literature, we were able to make several clear conclusions regarding the strength of our proposed lightweight hybrid framework. Table 7 shows the performance of the proposed lightweight multimodal framework when compared to existing work done in the field of DIQA

On the SmartDoc-QA dataset (Nayef et al., 2015 (1)), the proposed framework achieves a Macro F1-score of 0.9154 indicating strong and consistent detection of the document image defects under real-world conditions. When we compare this proposed model with methods evaluated on the same dataset, the advantage becomes clear as to why our framework is robust and effective. For example, the Siamese DIQA model by Peng and Wang, (2020) (20), while OCR-based IQA methods evaluated by Kopytek et al., (2024) (4) reports significantly lower correlation. Although the evaluation metrics differ, these results collectively indicate that the proposed framework achieves comparable or stronger performance while have a computationally lightweight architecture.

A key observation in the literature review is that many models were trained on synthetic dataset experience a significant drop in performance when they were evaluated on real-world datasets, for example, the Attention-based RNN proposed by Li, Pengchao et al. (2017) (8). In contrast, our proposed framework is trained and evaluated on real-world images, indicating that the model is not overfitting to controlled conditions and is capable of learning patterns that generalize to real-world defects.

Recently, high-performing approaches such as AHIQ (Lao et al., 2022 (17)) and Q-Doc (Huang et al., 2025 [12]) achieve strong results using transformer-based architectures and large language models were introduced. However, these models are computationally very expensive and require high performance and robust GPUs with other significant resources for deployment in any environment In contrast, the proposed framework achieves competitive performance with a model size of only 1.44MB demonstrating that strong real-world frameworks can be achieved without heavy computational requirements.

Table 7
Performance comparison of existing work with the proposed framework on DIQA tasks.

Ref & Paper	Model Used	Method Summary	Evaluation Metric	Result
[Proposed]	**CNN + Numerical**	**Multimodal fusion with balanced non-linear projection**	**Macro F1**	**0.9154**
Huang et al. (2020)	MobileNet	Video frame filtering for blur rejection	Accuracy	72.5%
Li et al. (2017)	CNN + SVR	Predicts single image quality score	LCC	~0.76
Li et al. (2017)	CNN + GRU + Attn	Sequential glimpse-based quality estimation	Accuracy	81%
Peng & Wang (2020)	Twin ResNet	Siamese comparison of image patches	PLCC	0.914
Kopytek et al. (2024)	Pseudo-F / CM3	OCR-driven quality estimation	Correlation	0.6037
Gao et al. (2025)	MLLM	High-resolution LLM-based assessment	Weighted Corr.	0.8989
Lao et al. (2022)	ViT + CNN	Hybrid global and local feature fusion	PLCC	0.8284
Li et al. (2018)	Handcrafted	Edge-based sharpness measurement	Monotonicity	33.6%
Huang et al. (2025)	LLMs	Step-by-step reasoning for DIQA	Accuracy	68.89%

5. Conclusion

This paper addresses the limitations of traditional DIQA models, which typically depends on a single type of input modality and produce only a single overall quality score which makes them less useful in real-world applications. To overcome this, we proposed a framework to detect multiple types of document defects such as blur, document skew and lighting problems, providing more meaningful and interpretable outputs

The proposed model efficiently combines visual features learned from images using a lightweight CNN with numerical features that describe the statistical properties of images, to help overcome the limitations of using either modalities independently. We also addressed an important issue of feature imbalance by introducing a projection based non-linear fusion strategy that balances information from both the modalities and learn complex patterns which improved the overall performance.

The results show that the proposed framework achieves a Macro F1-score of 0.9154 on the SmartDoc-QA dataset outperforming baseline models and showing competitive performance when compared to existing approaches. The model also shows more stable behaviour during training and validation when compared to pre-trained models which suffer from model overfitting.

Finally, we demonstrated that the proposed model is lightweight, efficient and suitable for real-world use, with a small model size (1.44 MB) and low processing time (0.0317 seconds per image), making it practical for real-time and low resource applications. Overall, this work shows that combining complementary features in a carefully designed multimodal fusion approach can lead to a lightweight, robust framework which is practical in real-world document quality assessment systems.

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